Chapter 4: Inference and Hypothesis Testing

Statistics for Environmental Science

Mallory Barnes

Rendered October 16, 2026

This PDF was rendered on October 16, 2026. The book is updated during the semester; the latest version of this chapter is at https://malloryb.github.io/statistics_E538/04-Foundation_Inference.html.

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> *Data and data sets are not objective; they are creations of human design. We give numbers their voice, draw inferences from them, and define their meaning through our interpretations.*
>
> Kate Crawford

Inferential statistics is the branch of statistics concerned with drawing conclusions about populations or causal relationships from sample data. It provides the tools to answer questions like: did our manipulation actually cause a change, or could the pattern we observed have arisen by random chance alone?

This chapter does that in two parts. The first part builds the logic by simulation, with no formulas: we work out what chance alone can produce, then compare a real result against it. The rest of the chapter replaces the simulation with the standard hypothesis testing machinery, because once you can assume a known reference distribution you no longer have to rebuild the chance window from scratch every time.

What chance alone can produce

An experiment asks whether a manipulation causes a change in what we measure. Some manipulations are so overwhelming that no statistics are required to see them. Flip a light switch up and the light comes on; flip it down and it goes off, every time. You don't need a t-test to tell you that flipping the light switch is the cause of the change in brightness.

Nothing in environmental science behaves like this. Effects are real signals buried in noisy, variable measurements, where flipping the switch changes the outcome *on average*. Not in every instance, and not by the same amount every time.

Random chance can produce apparent differences between groups even when no real difference exists. We have already seen this with sampling: two samples from the same distribution come out differently. The same principle applies when we compare group means. This is a fundamental problem for inference.

Consider a scenario where we expect to find no real difference. A researcher surveys 10 caves for white-nose syndrome, a fungal disease that has killed millions of North American bats, capturing 20 bats at each cave and counting how many show visible signs of infection. Figure 1 lays out the design.

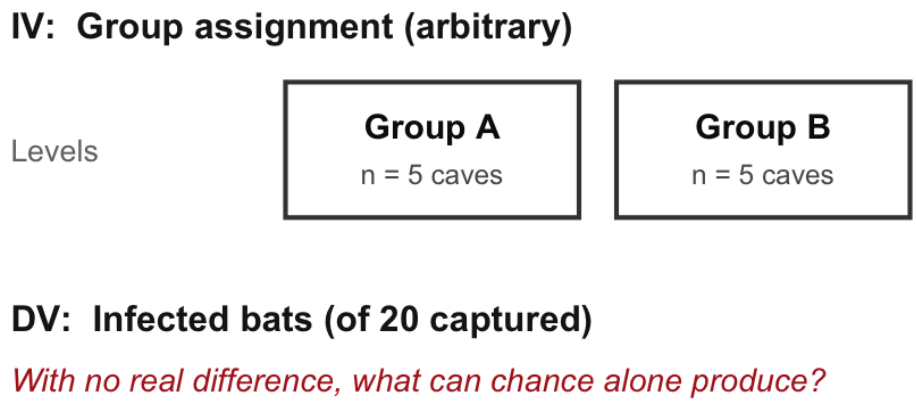


Figure 1: The bat-cave scenario study design. The split into two groups is arbitrary, which is the whole point: there is no real difference for the test to find.

The key feature is that all 10 caves are drawn from the same population. There is no treatment, no environmental gradient, nothing systematic separating them, so the true infection rate is identical everywhere. Splitting them into two groups of 5 is arbitrary. Any difference we see between those groups therefore *has* to be sampling error, which makes this the cleanest possible look at what chance alone can manufacture.

Figure 2 shows one run of this scenario: the mean number of infected bats in each group.

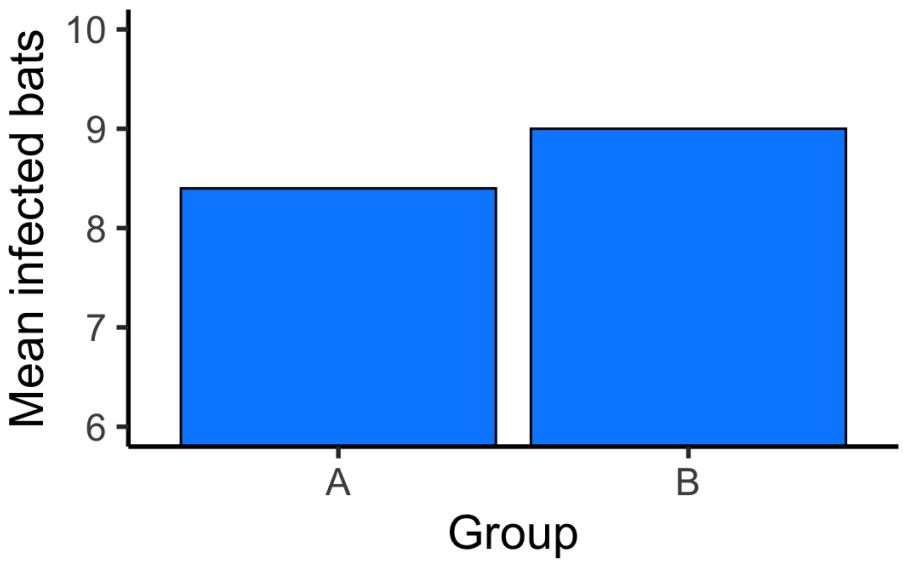


Figure 2: Mean number of infected bats per site for two groups drawn from the same population. Note the zoomed y-axis, which does not start at 0.

The bars are not the same height: Group A and Group B have different mean counts. But both groups were drawn from the very same population of sites, so this isn't a real difference in infection risk, it's sampling error.

This is the core inference problem: differences appear even when nothing real caused them. **How can we tell whether an observed difference is real or just the result of chance?**

Chapter 2 showed that chance alone can produce spurious correlations. The same is true for group means, and it can be pinned down: run the same arbitrary split many times and record what comes out. We summarize each run as a single **difference score**: the mean count for Group A minus the mean count for Group B. Figure 3 shows the difference scores for 10 replications of the survey, each one sampling all 10 caves from the same population, splitting them 5 and 5, and computing the means.

A bar at zero means the two groups had the same mean count in that replication. Positive values indicate Group A was higher; negative values indicate Group B was higher. Chance produces a new difference every time, but not an unlimited one. To see which sizes are common and which are rare, we run 10,000 replications and plot the difference scores as a histogram.

This histogram is the **chance window** for this study design. It shows that chance most often produces a difference near zero, the tallest bar in the center. Larger differences in either direction occur less and less frequently. Differences beyond about ± 3 are possible, but rare.

We can use this window to evaluate any observed difference. If a study found a difference of 1 infected bat between two groups, that falls well inside the chance window: differences of that size occur frequently by random sampling alone. If the study found a difference of 6 bats, that would fall far outside the window; chance produced a difference that large 0 times out of 10,000. When an observed result almost never happens by chance, we have grounds to conclude that *something other than chance* produced it.

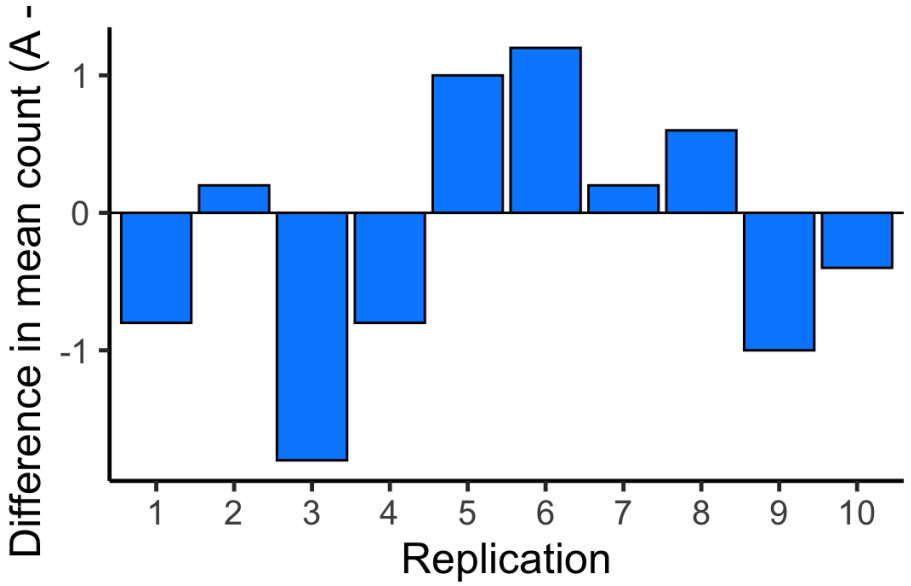


Figure 3: Difference in mean infected count between Group A and Group B across 10 replications. The true difference is zero, but sampling error produces non-zero differences each time.

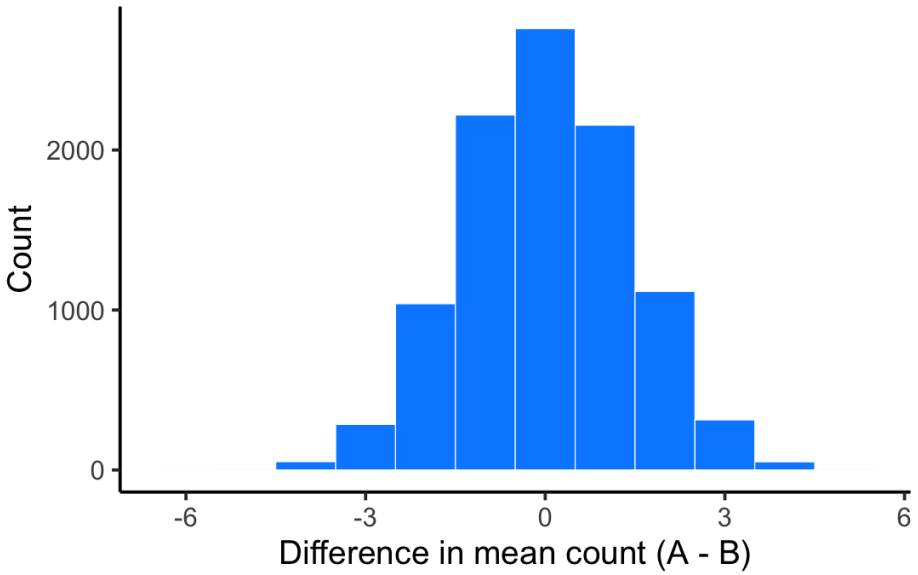


Figure 4: Histogram of mean count differences between Group A and Group B over 10,000 replications. All differences arise from chance alone. The most frequent difference is near zero; larger differences occur less often. Chance cannot produce everything.

Note: Which direction?

Still probability, model → data. The chance window starts from a world we chose, every cave at the same infection rate, and shows the differences that world produces. The 1-bat and 6-bat results above are hypothetical, used to show what the window can tell us. No real study's data have entered yet.

Chance vs. not chance

The window tells us what chance can do. It does not tell us where to draw the line between "chance could have" and "chance didn't." That is a judgment, not a calculation.

Figure 5 draws lines on the same bat histogram from above. The boundaries are the smallest and largest differences chance actually produced across the 10,000 replications.

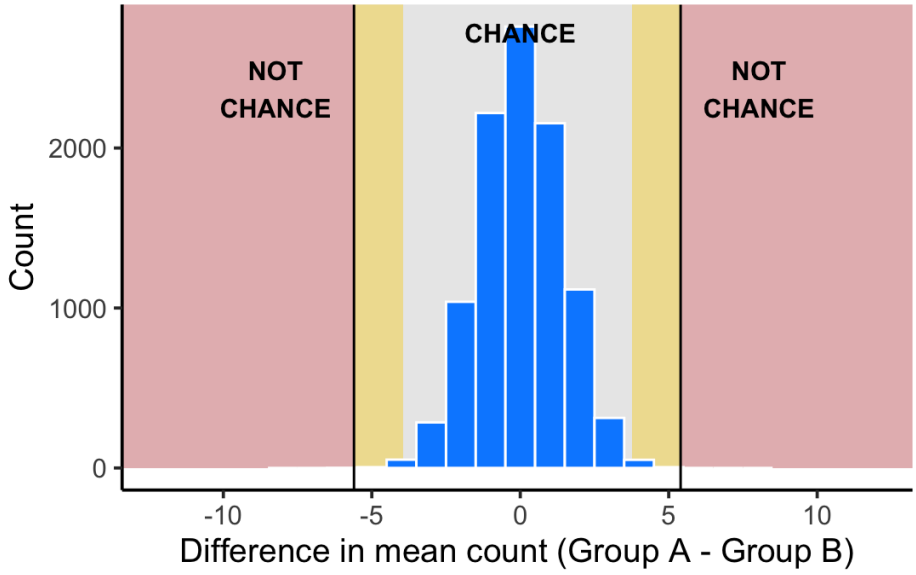


Figure 5: Decision boundaries applied to the histogram of mean count differences from Figure 4. Grey marks the range chance did produce across 10,000 replications; gold marks the edges of that range, which chance reached only rarely; red marks the differences chance never produced at all.

Across the 10,000 replications, chance produced differences from about -5.6 to 5.4 bats.

- **Chance (grey).** Chance produced differences like this routinely. No grounds for suspicion.
- **Ambiguous (gold).** The edges of what chance produced. Chance *can* land here, but rarely: probably not chance, but not ruled out.
- **Not chance (red).** Chance never produced a difference this large in 10,000 tries. Real grounds to suspect something beyond sampling error.

How rare is rare enough?

Question: How many times does something need to happen for it to happen "a lot"? How rare does something need to be before you stop worrying about it?

Environmental scientists already have a working vocabulary for this: the **100-year flood**. A "100-year flood" doesn't mean floods that size happen once a century on a schedule; it means that in any given year, there's about a 1-in-100 chance of a flood that severe. Would you build a house in that floodplain?

Many people do, and often regret it, because a 1% annual chance still adds up over a 30-year mortgage (roughly a 26% chance of at least one such flood in that span). A location with a 1-in-10,000 annual flood risk is very different: over the same 30 years, the cumulative chance is under 0.3%.

That's the intuition we need going forward: **rare events still happen, but how rare something is changes what you're willing to conclude from seeing it.**

Where the gold zone ends is a judgment call. A stricter line means fewer false alarms and more missed real effects; a looser line means the reverse. Science standardizes that call with a convention, $\alpha = .05$, defined later in this chapter. It is about 1 in 20, far looser than the red zone's never in 10,000.

Hypothesis tests

The null and alternative hypotheses

The chance window described a world where nothing but chance is at work. A formal test gives that world a name. In the bat example, "nothing but chance" meant no real difference between groups. From here on we compare a single sample against a specific value, such as a regulatory limit or a historical average.

The null hypothesis, H_0, is the claim that the population mean equals that specific value, written μ_0. If H_0 is true, any gap between the sample mean and μ_0 is sampling error.

The alternative hypothesis, H_a, is the claim that the population mean does not equal μ_0. It can also point in one direction only, larger or smaller; Section covers when that is justified.

$$H_0 : \ \mu = \mu_0 \qquad H_a : \ \mu \neq \mu_0$$

Null value notation. In this chapter, hypotheses are about the population mean μ, compared with the constant μ_0. The sample mean $\bar{x}$ is never part of H_0 or H_a; it is the evidence we bring to test them.

From simulation to a formal test

Everything so far was built by brute force: thousands of replications, piled into a histogram, with hypothetical results laid against the pile. Nobody wants to rebuild a chance window for every question. Chapter 3 already supplies a shortcut: sample means have a predictable distribution of their own: centered on the population mean, with spread $\sigma/\sqrt{n}$, and close to normal once the sample is large enough (the Central Limit Theorem). That lets us replace the simulated chance window with a curve whose shape we already know. The logic does not change, only the bookkeeping does.

> **i** Note: Which curves have to be normal
>
> The tests in this chapter and the next two read their answers off a normal curve, or its close relative, the t curve. Two different reasons put a normal curve there.
>
> - **For the sampling distribution of the sample mean**, it is the Central Limit Theorem at work: whatever the population looks like, sample means pile up in a roughly normal shape once the sample is large enough.
> - **For a population, it is not required.** A test reads its answer off the sampling distribution of

Table 1: The players in a hypothesis test, and the line each is drawn with in this chapter's figures.

	Line	Player	What it is
If H_0 were true		μ_0	The population mean if H_0 w
If H_0 were true		Null population distribution	The population if H_0 were tr
If H_0 were true		Sampling distribution of the sample mean under H_0	Where $\bar{x}$ would land across m
From the sample		$\bar{x}$	The sample mean: the one va
From the sample		Estimated population distribution	Centered at $\bar{x}$, with the samp
From the sample		Sampling distribution of the sample mean	Where $\bar{x}$ would land across m
From the sample		Confidence interval	$\bar{x} \pm 1.96$ standard errors, buil
The truth		True population distribution	The population the data actu

the sample mean, not off the population. The population's shape only matters when the sample is small, because the Central Limit Theorem needs a large enough sample to promise a normal shape for sample means. With a small sample, the normal curve for $\bar{x}$ rests on the population itself being roughly normal, and the only check we have is the sample: a histogram with no strong skew and no extreme outliers.

When the sample says the normal assumption is not reasonable, the nonparametric tests in Chapter 7 take over.

The players

A test is about μ, the true population mean, which we never observe; everything we know about it comes from one sample. The null hypothesis proposes a value for it, μ_0. To reject H_0, the sample has to be hard to explain if the null were true: its mean, $\bar{x}$, has to land farther from μ_0 than sampling error would plausibly carry it.

Several distributions and values take part in that comparison. Table 1 lists every one, with the line each is drawn with in this chapter's figures. They fall into three groups: what we would expect if H_0 were true, what the sample gives us, and the truth we never see.

Seeing them together

The players are easier to hold onto as a picture than as a table. Let's build that picture a curve or two at a time.

The population curves in these figures are drawn as normal for clarity; the note in the previous section explains why a test does not need them to be.

Notation reminder. Chapter 3's shorthand for a normal curve: $X \sim N(\mu, \sigma^2)$ reads "the random variable X is normally distributed with mean μ and variance σ^2." A normal curve needs just those two numbers, a center and a spread. The second number is the variance, the square of the standard deviation.

1. The null population distribution.

Start with the null population distribution: the population we would have if H_0 were true. Its center, μ_0, comes from H_0 itself. Its spread is the population standard deviation, σ. In practice σ is almost never known

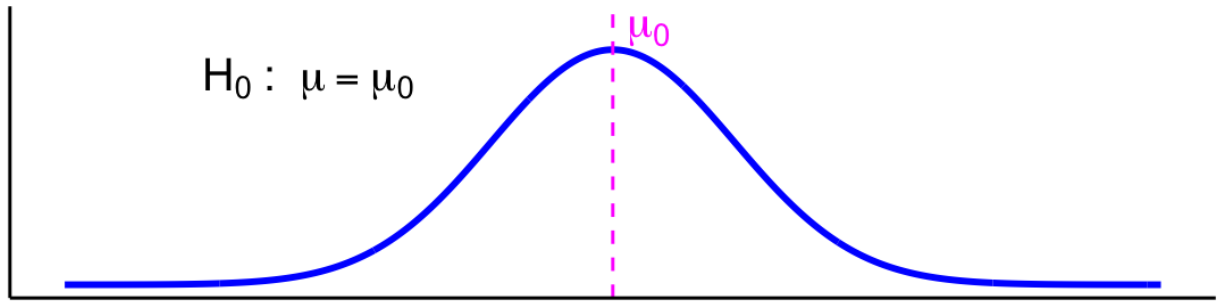


Figure 6: Null population distribution: the population we would have if H_0 were true, centered at μ_0.

and the sample's standard deviation stands in for it; this chapter treats σ as known as a simplification. The t-tests in Chapter 6 are the adjustment used in practice.

2. Null vs. true population.

In an idealized world where H_0 is exactly true, the true population and the null population distribution are identical. In reality, we never know the true population.

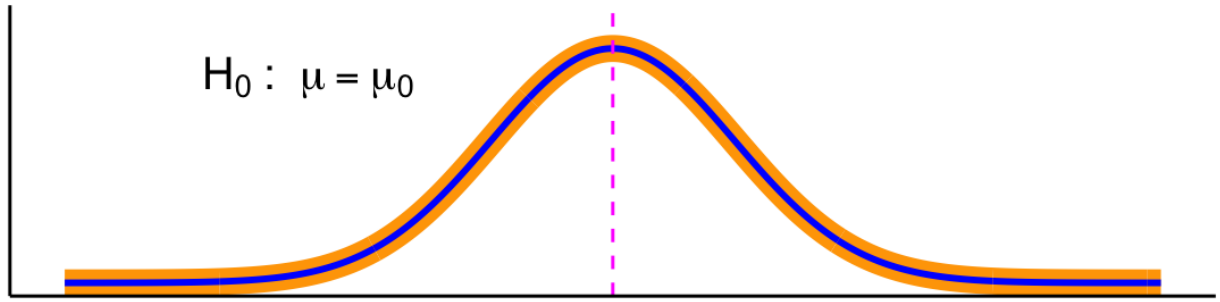


Figure 7: True population distribution equals the null population distribution (H_0 true): $\mu = \mu_0$.

> **i** Note: Which direction?
>
> **The switch to statistics, data → model.** Steps 1 and 2 came from H_0, a model we proposed. Step 3 starts from the sample we actually have and works back toward the population it came from.

3. The sample's picture of the truth. We never see the true population, but we do have a sample. From the sample's mean, $\bar{x}$, and its standard deviation we build the **estimated population distribution**: our best picture of the true population, drawn here as a normal curve. With a representative sample, the two line up closely.

4. From individual readings to sample means. By the Central Limit Theorem, the **sampling distribution of the sample mean** is close to a normal curve once the sample is large enough, so it needs only a center and a spread. Its center is $\bar{x}$, the same as the estimated population. Its spread is the standard error: the sample's standard deviation divided by $\sqrt{n}$. (Not $\sqrt{n-1}$: the $n-1$ already went into computing the standard deviation itself, back in Chapter 3.) In the shorthand, it is $N(\bar{x}, SE_{\bar{x}}^2)$: same center, much narrower. Figure 9 shows the Central Limit Theorem at work: make the population right-skewed, and the sampling distribution of the sample mean keeps the same normal shape and the same spread.

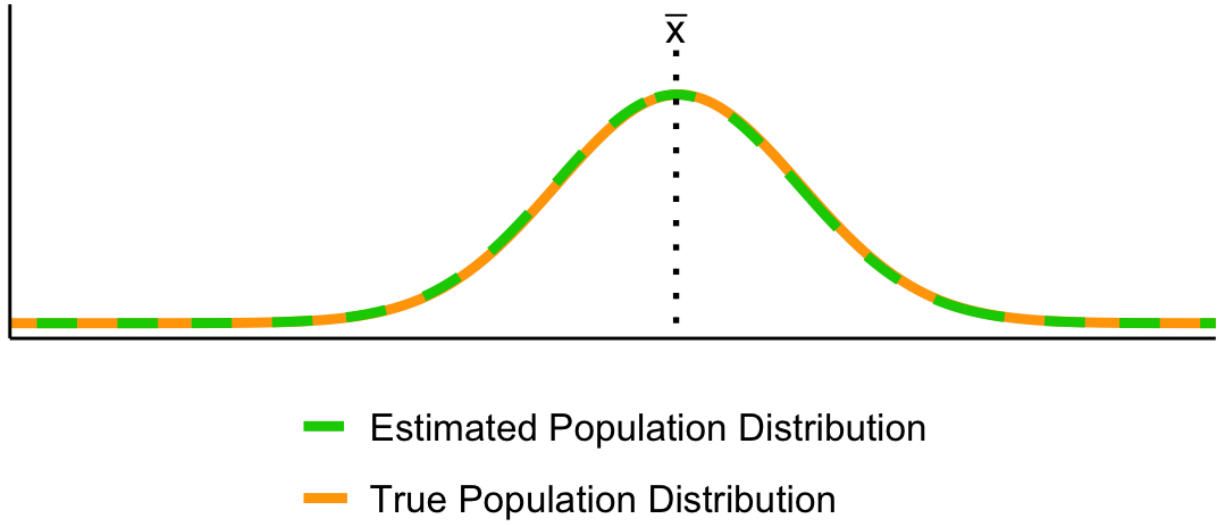


Figure 8: A representative sample: the estimated population distribution, built from the sample, lines up closely with the true population distribution.

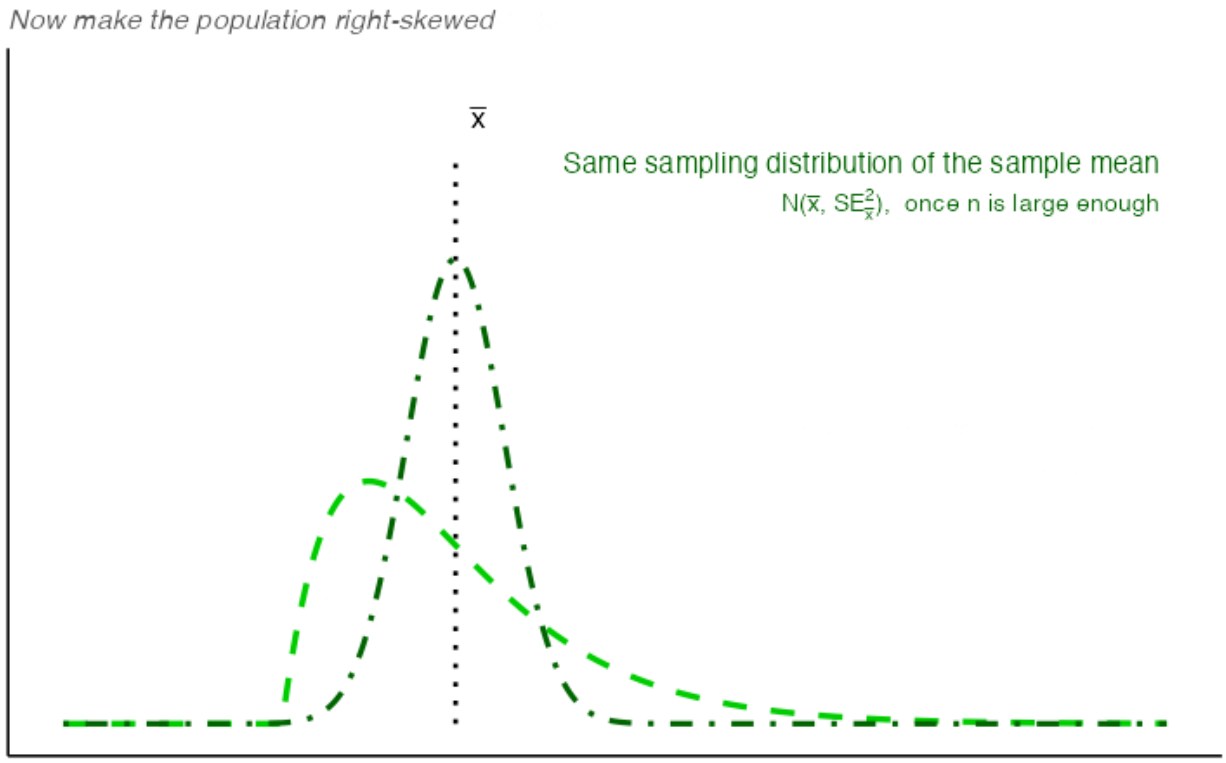


Figure 9: From individual readings to sample means. The sampling distribution of the sample mean (dark-green dot-dash) shares the estimated population's center, $\bar{x}$, with the standard error as its spread. When the population turns right-skewed, the sampling distribution keeps the same normal shape and spread, once the sample is large enough. (Final frame. The animated version is in the web edition of this book.)

5. The same move on the null side. The **sampling distribution of the sample mean under** H_0 is centered at μ_0, and its spread is again the standard error, the null population's standard deviation divided by $\sqrt{n}$ (with σ treated as known, the same standard error as in step 4): $N(\mu_0, SE_{\bar{x}}^2)$. It describes where $\bar{x}$ would land across many samples if H_0 were true, which makes it the formal version of the chance window. The critical values later in the chapter come from it.

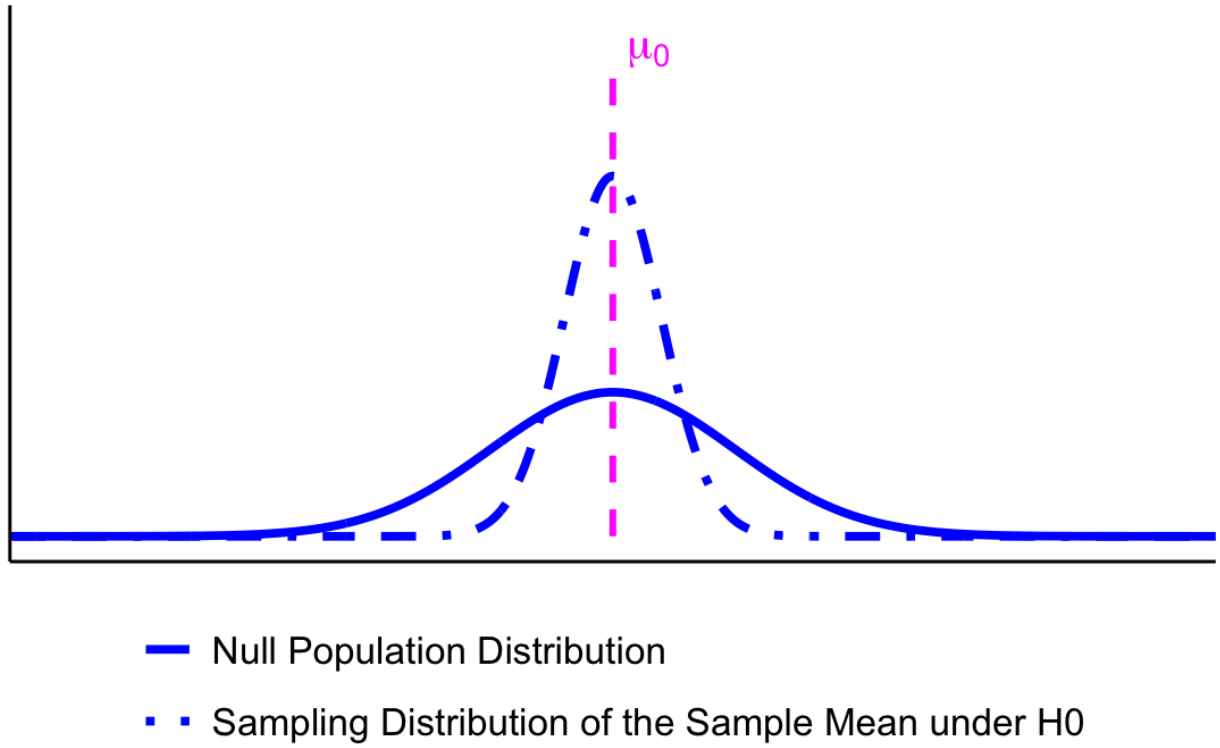


Figure 10: The same move on the null side: the null population distribution (blue solid) and the sampling distribution of the sample mean under H_0 (blue dot-dash) share a center, μ_0. Dividing the spread by $\sqrt{n}$ makes the second much narrower.

All of them at once. Put every player on one set of axes and the picture gets crowded fast, which is why the rest of the book shows only the pieces a given question needs. Figure 11 adds them one at a time, grouped as in Table 1, and ends with all of them together.

The two curves a decision uses. Strip the picture back to the two sampling distributions of the sample mean (Figure 12). Everything in the next section runs on these two: critical values and the p-value read off the one built from H_0, and the confidence interval comes from the one built from the sample.

Making the decision

Terms for decision rules

When running a hypothesis test, four terms guide our decision. The first is the quantity every test computes.

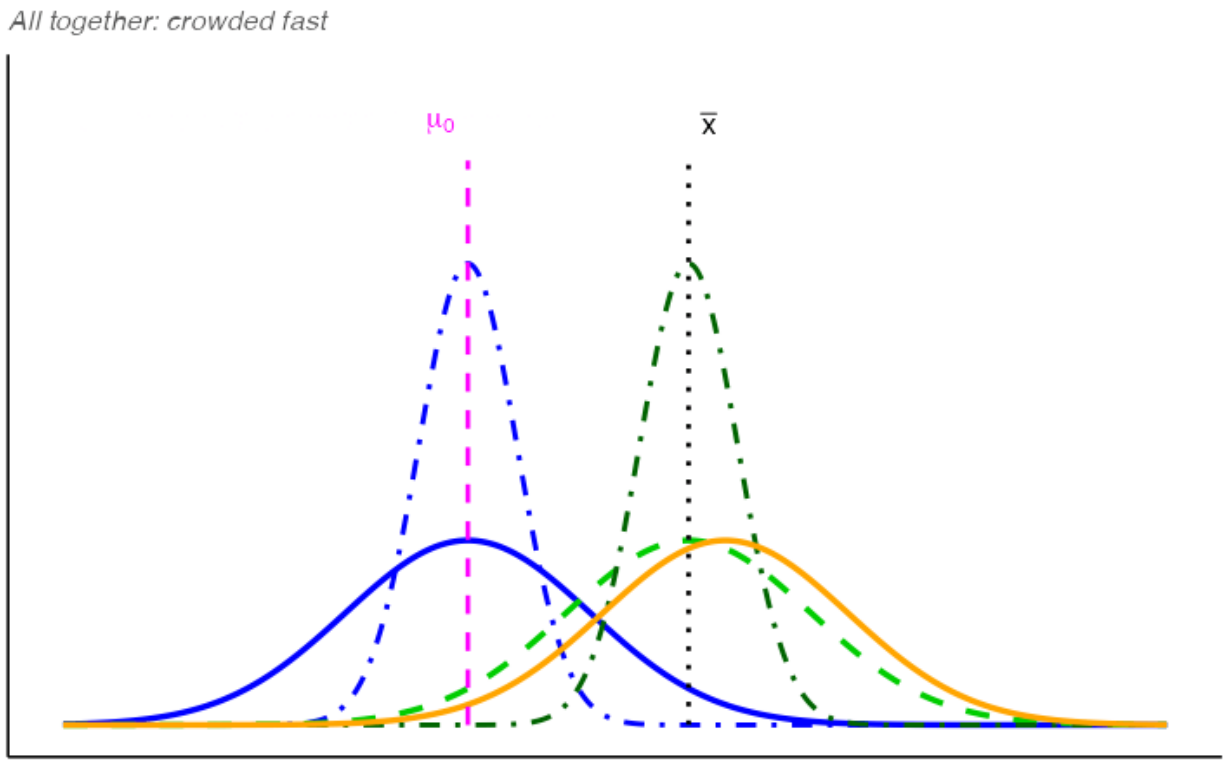


Figure 11: The players, added one at a time: first everything built from H_0, then everything built from the sample, then the true population, ending with all of them on one set of axes. (Final frame. The animated version is in the web edition of this book.)

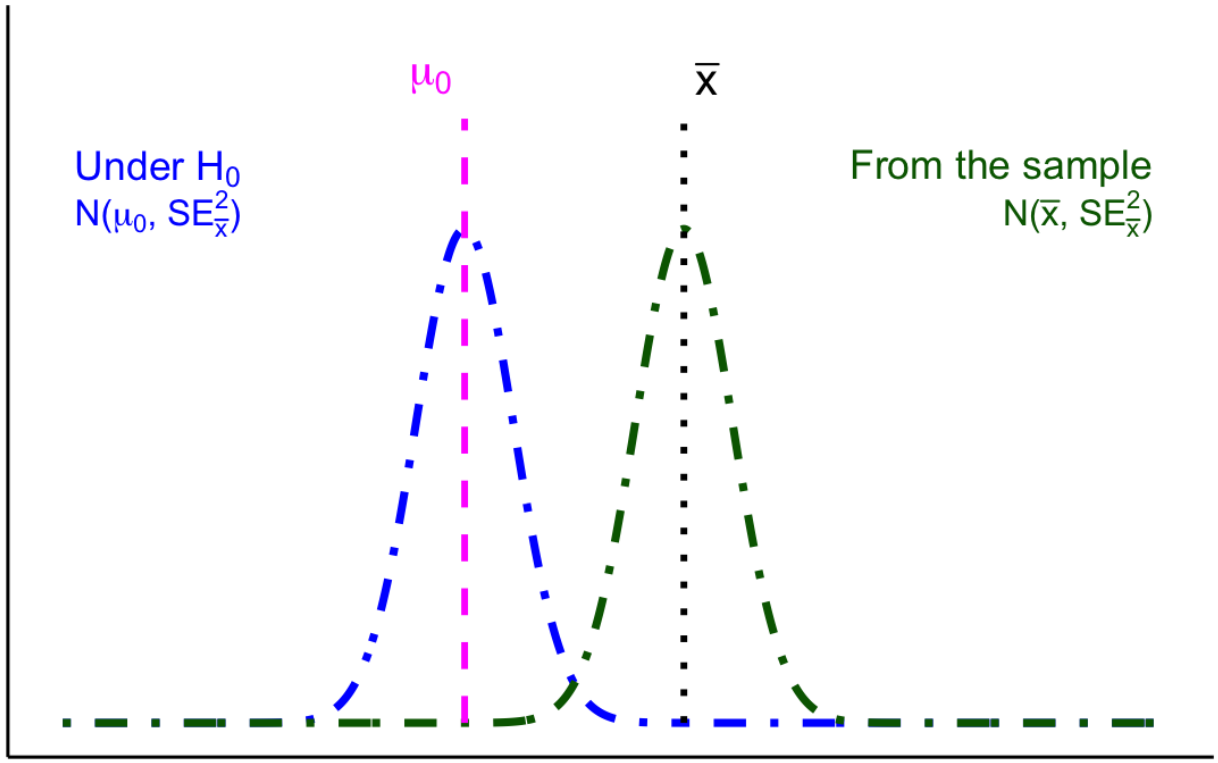


Figure 12: The two curves a decision uses: the sampling distribution of the sample mean under H_0 (blue dot-dash), centered at μ_0, and the sampling distribution of the sample mean from the sample (dark-green dot-dash), centered at $\bar{x}$. Both have the standard error as their spread.

Test statistic. A test statistic is a ratio. On top goes the effect you observed, the distance between what you measured and what H_0 predicted. On the bottom goes an estimate of how much that distance could move around by chance alone:

$$\text{statistic} = \frac{\text{effect}}{\text{error}}$$

That is the whole idea, and every test in this book is a version of it. A difference of 2 units is not impressive or unimpressive on its own. It is impressive only relative to how much difference chance produces for a study of this size, which is exactly what the chance window measured earlier in this chapter. The test statistic is that comparison written as a single number: how many units of ordinary noise is this effect worth?

Large statistic means the effect is big compared to the noise. Small statistic means it is not. Everything after this is bookkeeping about where to draw the line.

Significance level (α). Chosen *before* the test. α is the long-run probability of a Type I error (rejecting a true H_0; more on error types next chapter). A common choice is $\alpha = 0.05$. Smaller α makes tests more cautious and produces wider confidence intervals.

p-value. Assuming H_0 is true, the p-value is the probability of seeing a test statistic at least as extreme as what we observed. Small p means our data look unusual under H_0.

Confidence level. Confidence level is the long-run proportion of confidence intervals that contain the true parameter when we repeat the entire data collection and interval procedure.

$$\text{Confidence Level} = 1 - \alpha$$

With $\alpha = 0.05$, about 95% of intervals constructed this way will capture the true value.

> Warning: Don't misread a confidence interval
>
> Do **not** say:
>
> "There is a 95% probability this interval contains μ."
>
> Instead say:
>
> "We used a procedure that captures μ about 95% of the time in the long run."

Three equivalent ways to test H_0

All three routes below run on the same quantity: the **test statistic** defined above.

1. **Test statistic / critical values.** Choose α, find the corresponding **critical value(s)** for your test statistic, and compute your observed statistic. Reject H_0 if the statistic falls in the **rejection region**.
2. **Confidence interval (two-sided).** Build a $(1-\alpha)$ CI for the parameter. Reject H_0 if μ_0 lies **outside** the CI; fail to reject if it lies **inside**.
3. **p-value.** Compute the p-value for your observed statistic under H_0 and compare to α. Reject H_0 if $p < \alpha$.

Note. Methods 1 and 2 yield a yes/no at α; method 3 yields a yes/no plus a graded measure of evidence (exact p).

Route 1: test statistic and critical values

Critical values come from the sampling distribution of the sample mean under H_0, built in Figure 10.

Figure 13 draws it for a two-tailed test at $\alpha = 0.05$, marked off in standard errors from μ_0. The red lines are the **critical values**, 1.96 standard errors on either side of μ_0. They cut off the most extreme 5% of the curve, 2.5% in each tail, and those shaded tails are the **rejection regions**. Here the observed $\bar{x}$ lands 2.4 standard errors above μ_0, inside the right-hand rejection region. Counting that distance in standard errors is exactly the test statistic from earlier, $z = (\bar{x} - \mu_0)/SE = 2.4$.

Here's the decision rule:

- If the observed test statistic falls inside the central region, we **fail to reject** H_0 because the result is consistent with chance variation.
- If it falls in either rejection region, we **reject** H_0 because the result is too extreme to attribute to chance alone at the 5% level.

This picture makes clear why α and the p-value are linked. α sets the cutoff for how extreme is "too extreme," while the p-value tells us how far into the tails our actual result lies.

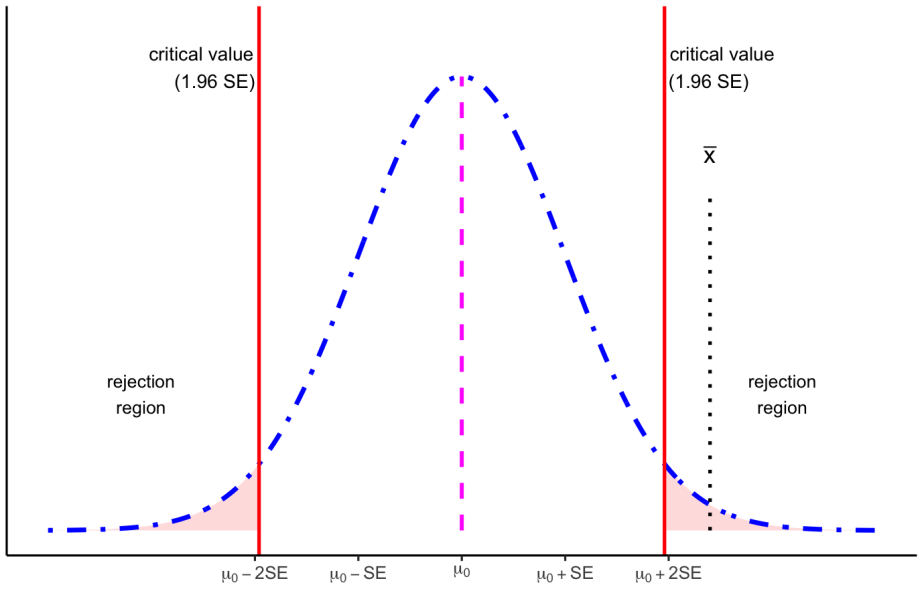


Figure 13: Two-tailed test at $\alpha = 0.05$: the sampling distribution of the sample mean under H_0 (blue dot-dash), centered at μ_0 (magenta) and marked off in standard errors. Red lines are the critical values at ± 1.96 SE; the shaded tails are the rejection regions. The observed $\bar{x}$ (dotted) falls 2.4 SE above μ_0, in the rejection region.

Route 2: confidence interval

The confidence-interval route makes the same decision from the sample's side. It never uses the null population distribution, only the single value μ_0.

Start from the sampling distribution of the sample mean built in Figure 9: centered at $\bar{x}$, with the standard error as its spread. A sample mean lands within 1.96 standard errors of the true mean μ about 95% of the time, so the middle 95% of this curve, $\bar{x} \pm 1.96\,SE$, is the 95% confidence interval: the values of μ that are consistent with our sample (Figure 14). Then ask where μ_0 falls (Figure 15).

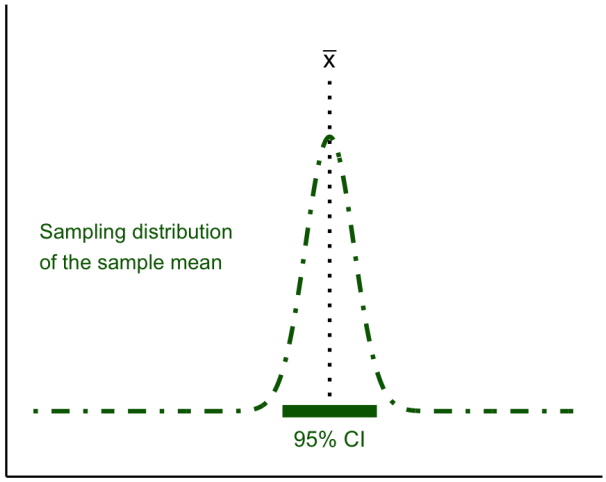


Figure 14: The 95% confidence interval: the middle 95% of the sampling distribution of the sample mean, $\bar{x} \pm 1.96$ standard errors.

Decision rule:

- If μ_0 lies **inside** the CI bar, we **fail to reject** H_0.
- If μ_0 lies **outside** the CI bar, we **reject** H_0 at $\alpha = 0.05$.

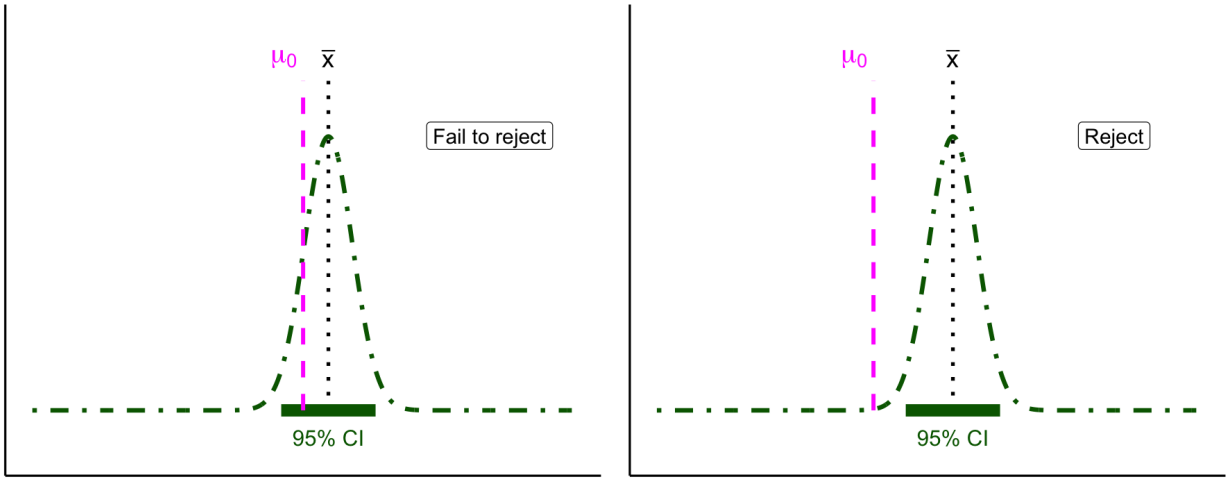


Figure 15: The decision, same sample in both panels. Left: μ_0 (magenta) falls inside the 95% confidence interval, so we fail to reject H_0. Right: a different μ_0 falls outside it, so we reject H_0.

The right panel is worth a second look. A single reading at that μ_0 would be unremarkable: it sits well inside the wide estimated population of Figure 9. It is simply not a plausible value for the *mean*, because sample means vary much less than individual readings do.

For common models, a two-tailed level-α test of $H_0 : \mu = \mu_0$ is equivalent to checking whether μ_0 lies inside the $100(1-\alpha)\%$ confidence interval. The **CI** is often the most intuitive way to see this: it shows both the decision (reject or not) *and* the range of parameter values still consistent with the data.

Route 3: *p*-value

The p-value is an area under the curve, so it comes from `pnorm()`, which Chapter 3 introduced as the area to the left of a value. For the observed $z = 1.85$ in Figure 16, `pnorm(-1.85)` gives the area below -1.85, about 0.032, which by symmetry equals the area above $+1.85$. A two-tailed test counts both tails, so $p =$ `2 * pnorm(-1.85)` ≈ 0.064. Reject if $p < \alpha$; here $0.064 > 0.05$, so we fail to reject.

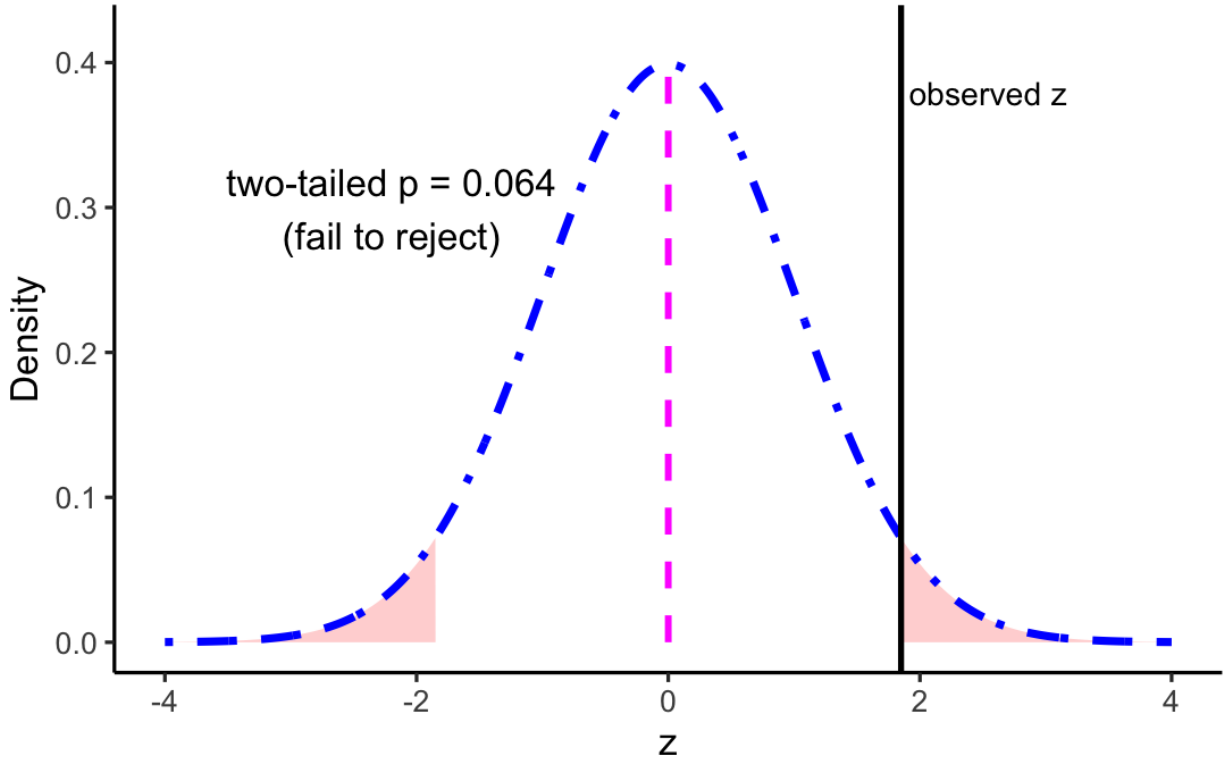


Figure 16: p-value lens: the two-tailed p-value is the shaded area beyond the observed statistic (vertical line), in both tails, under the null.

Reject or fail to reject

Test outcomes are **binary**: we either **reject** H_0 or **fail to reject** H_0. We never *accept* H_0, because sampling variability means we can't prove it's true, only that the data are consistent with it. Similarly, we don't formally *accept* H_a, but we can say the results provide **support for** H_a when H_0 is rejected.

Report the exact p and an effect size or CI.

Here are our options (for a two-tailed test).

1. **Reject the Null Hypothesis (Two-Tailed Test).** The sample mean falls beyond the critical values, in the rejection region. The confidence interval also excludes the null mean. In this example, $\bar{x}$ is 3.75 standard errors from μ_0 ($z = -3.75$), well past the critical values of ± 1.96, and the two-tailed p-value is about 0.0002, far below $\alpha = 0.05$. All three routes say reject.
2. **Fail to Reject the Null Hypothesis (Two-Tailed Test).** The sample mean falls in the non-rejection region, and the confidence interval includes the null mean. Here $\bar{x}$ is 1 standard error from μ_0 ($z = -1.00$), inside the critical values, and the two-tailed p-value is 0.32, above $\alpha = 0.05$. All three routes say the evidence is consistent with chance variation, so we fail to reject H_0.

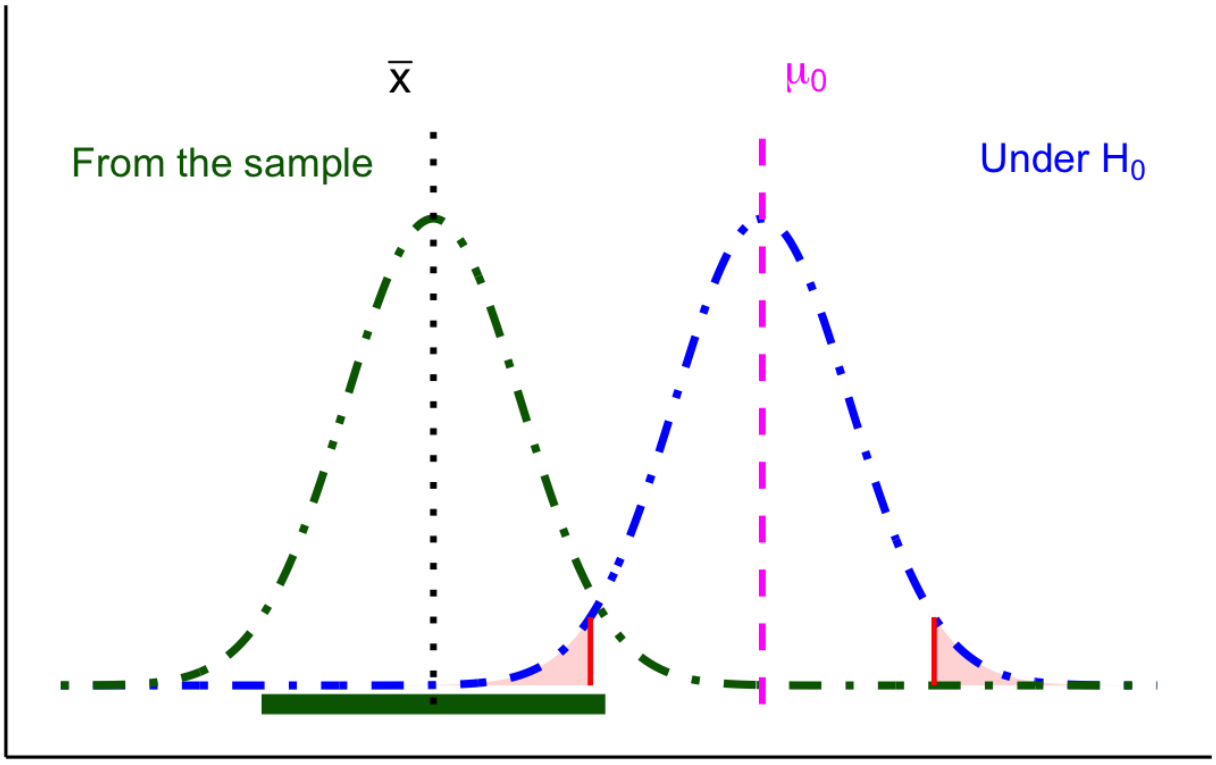


Figure 17: Two-tailed test, reject H_0. Blue: the sampling distribution of the sample mean under H_0, with its rejection regions shaded beyond ± 1.96 SE; $\bar{x}$ lands in one. Green: the sampling distribution of the sample mean from the sample, with its 95% confidence interval; μ_0 falls outside it.

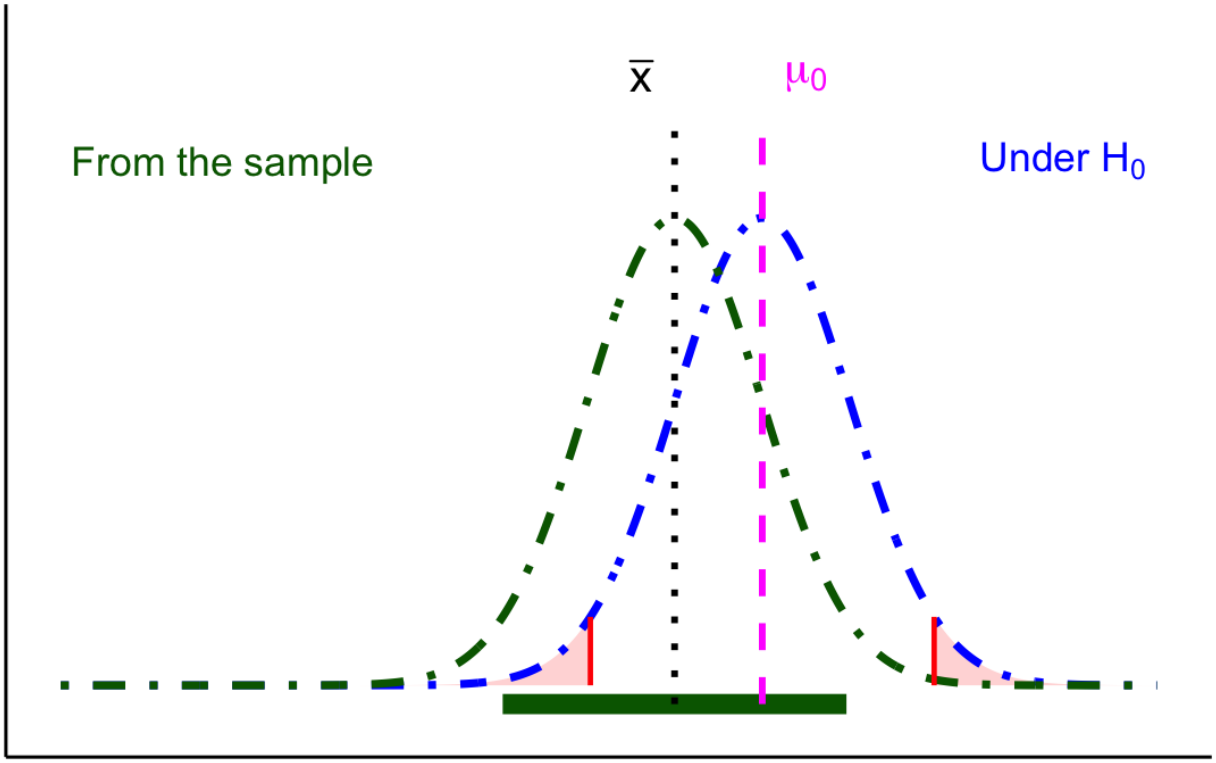


Figure 18: Two-tailed test, fail to reject H_0. $\bar{x}$ lands 1 standard error from μ_0, inside the critical values of ± 1.96 SE, and the confidence interval from the sample includes μ_0.

> Warning: Beyond ruling out chance
>
> Every hypothesis test in this book asks one question: could chance alone have produced this result? It is not the only question available.
> Other methods start from two or more candidate explanations, say a steady linear decline versus a sharp threshold response, and ask which one the data support better.
> Ruling out chance tells you that *something* is going on, but not always *what*. It is worth knowing the difference, because "significant" only ever answers the first question.

One- and two-tailed tests

H_a is a claim about where the true population could sit. A two-tailed alternative says it might sit below μ_0 or above it. A one-tailed alternative says it can only sit on one side (Figure 19).

The test handles those claims through where α goes. A two-tailed test splits α between both tails of the sampling distribution of the sample mean under H_0. A one-tailed test puts all of α in the one tail H_a points toward (Figure 20).

One- or two-tailed: which and why?

Use this rule of thumb:

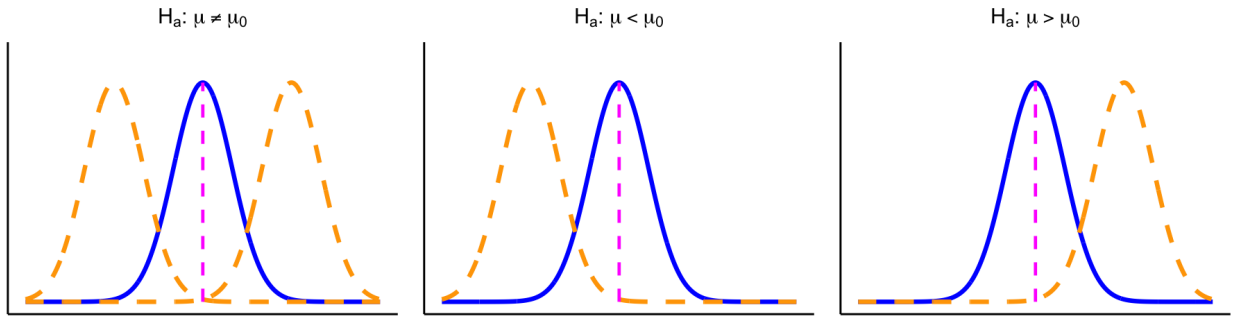


Figure 19: What H_a claims. The null population distribution (blue) sits at μ_0. The orange dashed curves are possible true population distributions under H_a: on either side for a two-tailed alternative (left), on one side only for a one-tailed alternative (middle and right).

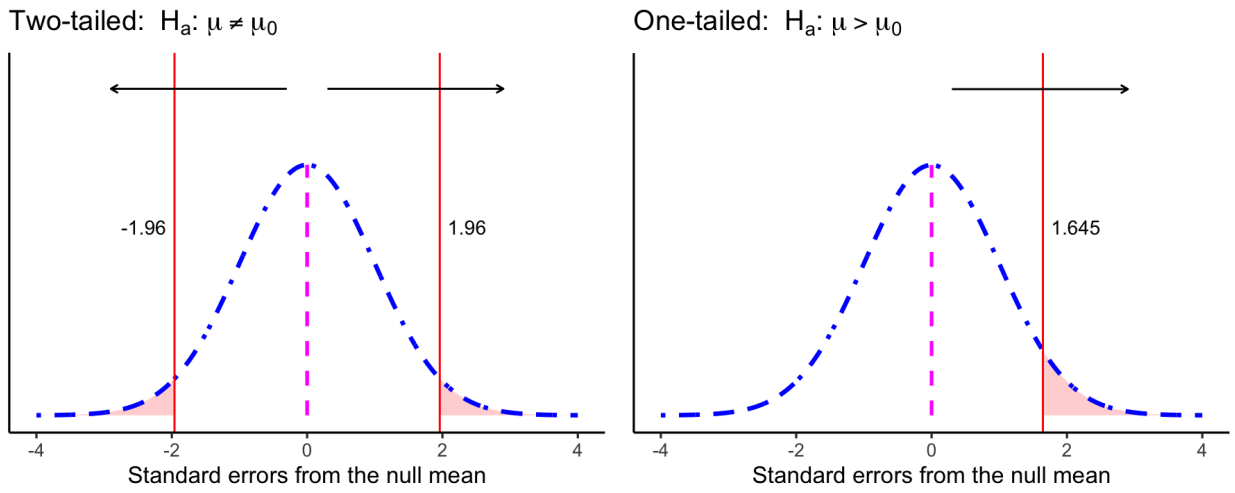


Figure 20: Where α goes. Both panels show the sampling distribution of the sample mean under H_0 (blue dot-dash), in standard-error units. Left, two-tailed ($H_a: \mu \neq \mu_0$): $\alpha/2$ in each tail, critical values at ± 1.96. Right, one-tailed ($H_a: \mu > \mu_0$): all of α in the right tail, critical value at 1.645.

- Two-tailed (default): You care whether μ differs from μ_0 **in either direction**.
- One-tailed (only if justified): Differences in the opposite direction are **scientifically irrelevant** and **would not change your decision or action**.

Default to two-tailed unless your theory rules out the other direction. At $\alpha = .05$, a one-tailed test's critical value is 1.645 instead of 1.96, so it rejects on less evidence, but only in the direction you named in advance. If the effect goes the other way, a one-tailed test cannot detect it at all.

That is why the direction has to be chosen before you see the data. Picking the tail afterward, because it gives the smaller p-value, makes your real chance of a false positive higher than the α you report.

Chapter Summary

Why it matters. This chapter is the bridge between describing data and drawing conclusions from it. Everything after this point is a variation on one move: work out what chance alone could produce, then ask whether your result is rare enough under chance, by a standard set in advance (α), to rule chance out.

Core ideas

- **Chance produces differences on its own.** Split one population into two arbitrary groups and their means will differ anyway, from sampling error alone. The question is never "is there a difference," it is "is this difference bigger than chance can account for."
- **The chance window is buildable.** Repeat a study thousands of times under conditions where you know nothing real is going on, collect the differences, and you have a picture of exactly what chance is capable of for a design your size.
- **Rarity is a judgment, and it needs a convention.** The 100-year flood shows the reasoning already exists outside statistics: a 1% annual chance is not zero, but it changes what you build. $\alpha = .05$ is that same judgment, standardized so researchers aren't each drawing their own line.
- **A hypothesis test is the practical shortcut.** Once you can assume a known reference distribution, the simulated chance window becomes a curve, and the decision becomes a calculation. The logic is the same.
- **Two pairs of curves.** From the sample: the estimated population distribution and the sampling distribution of the sample mean, which gives the confidence interval. Under H_0: the null population distribution and the sampling distribution of the sample mean under H_0, which gives the critical values and p-value. The true population distribution is never known.
- **Three routes, one decision.** A test statistic against a critical value, a confidence interval, and a p-value are three ways of asking the same question. For a two-tailed test at the same α they always agree; a one-tailed test checked against a two-sided confidence interval can disagree near the boundary.
- **Commit to one- or two-tailed in advance.** A one-tailed test is only justified when theory rules out the other direction. Choosing after seeing the data raises your real error rate above the α you claimed.